Home SIEM Lab Setup & SSH Brute Force Detection

Technical Documentation

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Platform: Splunk Enterprise 10.2.1 (Free License)

Environment: Windows Host + Kali Linux VM (VirtualBox)

1. Project Overview

This document details the setup of a home Security Information and Event Management (SIEM) lab using Splunk Enterprise, and the execution of a controlled SSH brute force attack simulation to validate the detection capabilities of the SIEM. The lab was built on a Windows host machine running a Kali Linux virtual machine via Oracle VirtualBox.

1.1 Objectives

- Deploy Splunk Enterprise on a Windows host as the central SIEM platform
- Install and configure the Splunk Universal Forwarder on Kali Linux to forward logs
- Enable SSH on the Kali VM to serve as a target for testing
- Simulate an SSH brute force attack using Hydra on the Kali machine
- Verify that Splunk detects and displays the failed login attempts in real time

1.2 Lab Architecture

Component	Role	Details
Windows Host	SIEM Server	Splunk Enterprise 10.2.1, Free License, port 8000
Kali Linux VM	Monitored Endpoint / Attacker	VirtualBox VM, Splunk Universal Forwarder 9.3.2
Splunk Forwarder	Log Shipper	Forwards logs from Kali to Windows on port 9997
Hydra	Brute Force Tool	Pre-installed on Kali, used for SSH password attacks
OpenSSH	Attack Target	SSH server running on Kali port 22

2. Splunk Enterprise Setup on Windows

Splunk Enterprise was installed on the Windows host machine to serve as the central log aggregation and analysis platform. The free license tier was used, which provides up to 500MB of data ingestion per day sufficient for a home lab environment.

2.1 Installation

The Splunk Enterprise installer was downloaded from splunk.com as a Windows .msi package. Installation was performed with Administrator privileges using default settings. Splunk runs as a background Windows service and is accessible via a web browser.

Key Installation Details

Parameter	Value
Install Path	C:\Program Files\Splunk
Web Interface	http://localhost:8000
Admin Username	Whitebeard
Service Account	Local System
License Type	Free (500MB/day)
Splunk Version	10.2.1

2.2 Configuring a Receiving Port

To allow the Kali Linux Universal Forwarder to send logs to Splunk, a receiving port was configured in Splunk Web. This port listens for incoming log data from forwarding agents.

Navigation path in Splunk Web:

- Settings > Forwarding and Receiving > Configure Receiving > New Receiving Port
- Port configured: 9997

3. Splunk Universal Forwarder on Kali Linux

The Splunk Universal Forwarder is a lightweight agent that collects and ships logs from the Kali Linux VM to the Splunk Enterprise instance on Windows. Version 9.3.2 was installed.

3.1 Installation

The Universal Forwarder .deb package was downloaded directly from Splunk's servers and installed using the Debian package manager.

Download command:

```
wget -O splunkforwarder.deb
"https://download.splunk.com/products/universalforwarder/releases/9.3.2/li
nux/splunkforwarder-9.3.2-d8bb32809498-linux-2.6-amd64.deb"
```

Installation command:

```
sudo dpkg -i splunkforwarder.deb
```

First-time startup and license acceptance:

```
sudo /opt/splunkforwarder/bin/splunk start --accept-license
```

During startup, administrator credentials were created for the forwarder:

- Username: admin
- Password: [set during installation]

3.2 Connecting the Forwarder to Splunk

The forwarder was pointed to the Windows host IP address on port 9997 to establish the log forwarding connection.

```
sudo /opt/splunkforwarder/bin/splunk add forward-server
10.190.133.138:9997
```

Successful output:

```
Added forwarding to: 10.190.133.138:9997.
```

3.3 Configuring Log Sources

Kali Linux uses systemd journald instead of traditional syslog files. Since journal logs are stored in binary format, a dedicated text log file was created to allow Splunk to read them.

Creating a persistent journal log directory:

```
sudo mkdir -p /var/log/journal
sudo systemd-tmpfiles --create --prefix /var/log/journal
sudo systemctl restart systemd-journald
```

Exporting journal logs in real time to a readable file:

```
sudo journalctl -f --output=export | sudo tee /var/log/splunk-
journal/journal.log > /dev/null &
```

Adding the log sources to the forwarder:

```
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/boot.log
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/dpkg.log
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/wtmp
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/btmp
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/journal
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/splunk-
journal/journal.log
```

Restarting the forwarder to apply changes:

```
sudo /opt/splunkforwarder/bin/splunk restart
```

4. Verifying Log Ingestion in Splunk

After the forwarder was configured and restarted, log ingestion was verified in Splunk Web using the Search & Reporting app. The following SPL (Search Processing Language) query was used to confirm logs from the Kali host were being received.

4.1 Initial Verification Search

```
index=* host=kali
```

Result: Splunk returned events sourced from host=kali, including dpkg.log entries showing the Splunk forwarder installation history confirming that the log pipeline was working correctly end-to-end.

Field	Value
host	kali
source	/var/log/dpkg.log
sourcetype	dpkg-too-small
Events returned	Multiple pages (5+ pages of results)

5. SSH Brute Force Attack Simulation

With the SIEM operational and log ingestion confirmed, a controlled SSH brute force attack was simulated using Hydra a pre-installed password cracking tool on Kali Linux. The attack was performed locally on the Kali VM (loopback interface) against a dedicated test user account.

5.1 Prerequisites

Enabling SSH on Kali

The OpenSSH server was installed but disabled by default on Kali. It was started and enabled to persist across reboots.

```
sudo systemctl start ssh
sudo systemctl enable ssh
```

SSH was confirmed to be listening on port 22 across all interfaces:

```
ss -tnlp | grep 22
LISTEN 0   128   0.0.0.0:22   0.0.0.0:*
```

Creating a Test User Account

A dedicated non-privileged test account was created to serve as the brute force target, keeping the attack isolated from the main Kali user account.

```
sudo useradd -m testuser
sudo passwd testuser
```

Password set: kali (intentionally weak for testing purposes)

5.2 Executing the Brute Force Attack

Hydra was used to perform a dictionary-based SSH brute force attack against the testuser account on localhost. A custom wordlist of 11 common passwords was created, with the correct password included at the end to simulate a successful credential discovery.

Creating the Wordlist

```
cat << 'EOF' > /tmp/wordlist.txt
abc123
password
```

```
123456
admin
letmein
qwerty
welcome
monkey
dragon
master
kali
EOF
```

Running Hydra

```
hydra -l testuser -P /tmp/wordlist.txt ssh://127.0.0.1 -t 4 -v
```

Hydra command parameters:

Parameter	Description
-l testuser	Target username to attack
-P /tmp/wordlist.txt	Path to the password dictionary file
ssh://127.0.0.1	Protocol and target host (localhost loopback)
-t 4	Number of parallel threads
-v	Verbose mode — shows each attempt in real time

Attack Result

Hydra successfully identified the correct password after exhausting the wordlist:

```
[22][ssh] host: 127.0.0.1   login: testuser   password: kali
1 of 1 target successfully completed, 1 valid password found
```

This output demonstrates a complete brute force lifecycle multiple failed attempts followed by a successful credential discovery generating authentication failure events in the system logs.

6. Brute Force Detection in Splunk

After the Hydra attack completed, Splunk was queried to verify that the failed SSH login attempts were captured and visible in the SIEM. This validates the full detection pipeline from attack execution through log collection to SIEM visibility.

6.1 Splunk Search Query

```
index=* host=kali "Failed password"
```

Time range: Last 15 minutes

6.2 Detection Results

Splunk returned 33 events containing failed SSH authentication attempts. The events clearly showed the brute force pattern repeated failed logins from the same source IP against the same username in rapid succession.

Field	Value
Total Events	33
host	kali
source	/var/log/splunk-journal/journal.log
sourcetype	journal
Target User	testuser
Source IP	127.0.0.1
Protocol	SSH (port 22)
Attack Tool	Hydra v9.5

6.3 Sample Log Events

The following log entries were captured by Splunk from the journal log, showing the brute force pattern:

```
MESSAGE=Failed password for testuser from 127.0.0.1 port 60276 ssh2
MESSAGE=Failed password for testuser from 127.0.0.1 port 60274 ssh2
MESSAGE=Failed password for testuser from 127.0.0.1 port 60272 ssh2
MESSAGE=Accepted password for testuser from 127.0.0.1 port 49132 ssh2
```

These events represent the complete attack lifecycle captured in the SIEM from the initial failed attempts through to the final successful authentication, exactly as a SOC analyst would see it in a real environment.

7. Conclusions & Next Steps

7.1 What Was Achieved

- Successfully deployed Splunk Enterprise as a functional home SIEM on a Windows host
- Configured the Splunk Universal Forwarder on Kali Linux to ship logs in real time
- Validated the end-to-end log pipeline from Kali Linux to Splunk on Windows
- Executed a realistic SSH brute force attack using Hydra with a custom wordlist
- Confirmed Splunk detected and displayed 33 failed SSH login events from the attack
- Demonstrated the complete attacker/defender lab dynamic using a single Windows + VM setup

7.2 Recommended Next Steps

Next Step	Description	Priority
Create Alert Rules	Set up Splunk alerts that trigger when failed logins exceed a threshold (e.g., 5 in 1 minute)	High
Add Windows Event Logs	Install Splunk Add-on for Windows and ingest Security/System event logs from the host	High
Install Sysmon	Deploy Microsoft Sysmon on Windows for enhanced process, network, and file activity logging	Medium
BOTS Dataset	Load the Boss of the SOC (BOTS) free dataset for practicing real-world investigation scenarios	Medium
Security Essentials App	Install Splunk Security Essentials from Splunkbase for pre-built detection rules and dashboards	Low

8. Screenshots

The following screenshots are for evidence and documentation purposes

Figure 1.0 Port Forwarding Configuration

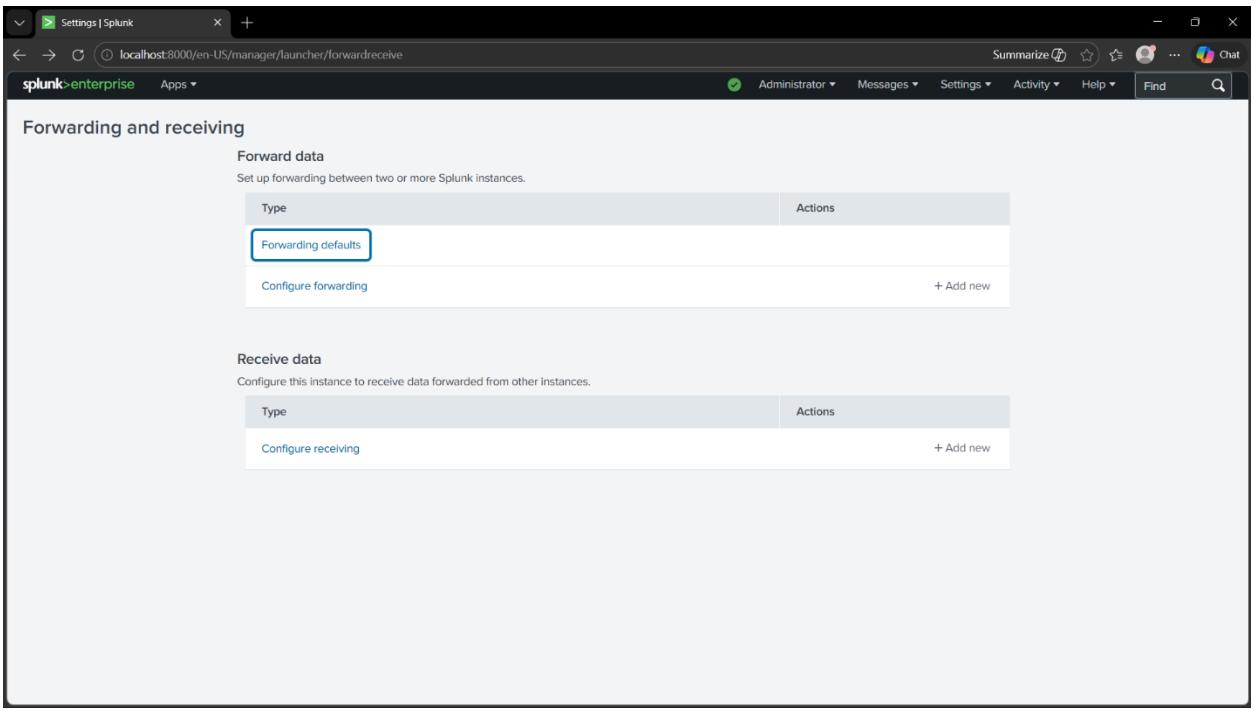

Figure 2.0 Splunk showing the dpkg.log events from Kali

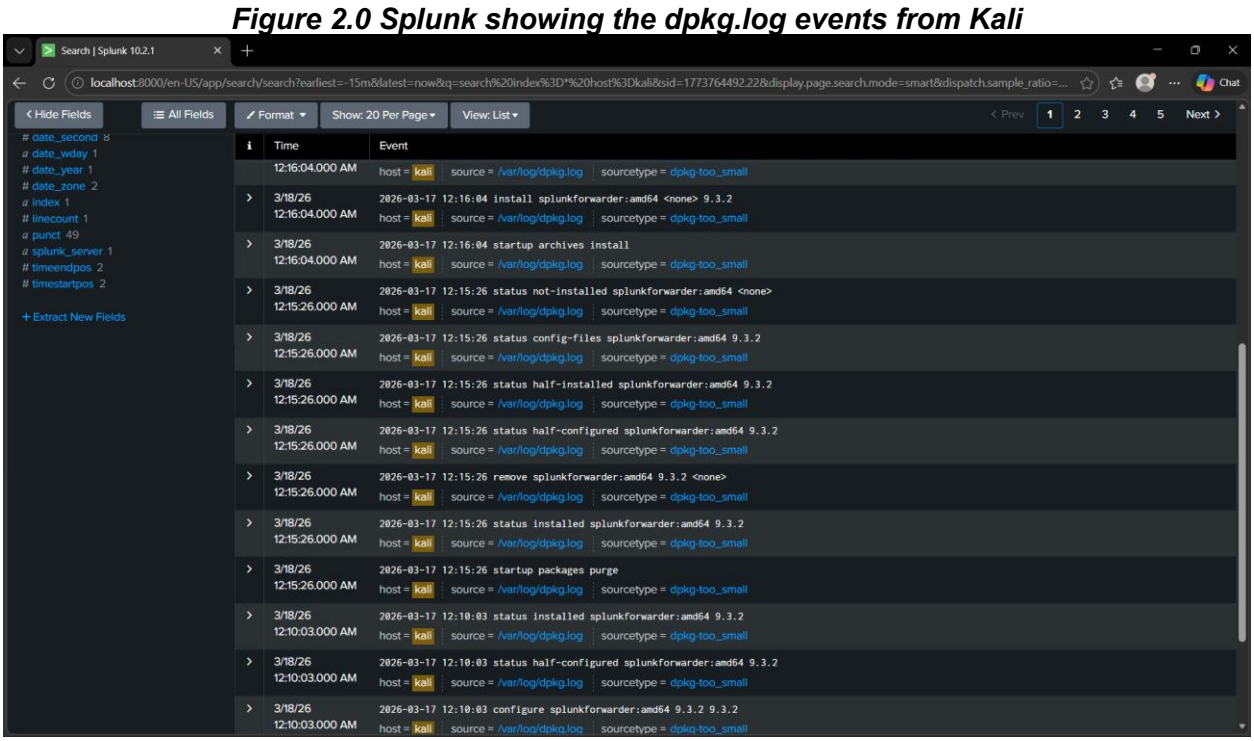

Figure 3.0 Reinstalling/Restarting the Splunk Universal Forwarder

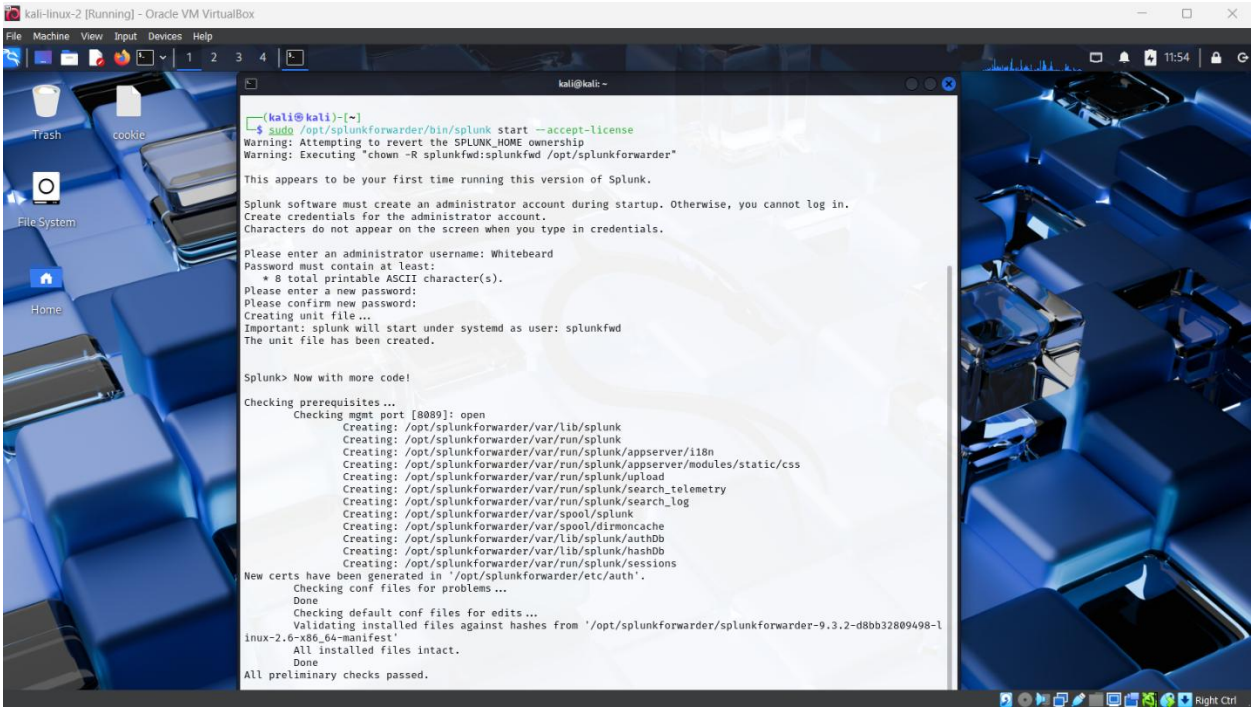

Figure 4.0 re-adding all the log monitors and restarting the forwarder

kali-linux-2 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

kali@kali: ~

```
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/boot.log'.
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/dpkg.log'.
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/wtmp'.
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/btmp'.

┌──(kali㉿kali)-[~]
└─$ sudo mkdir -p /var/log/journal
sudo systemd-tmpfiles --create --prefix /var/log/journal
sudo systemctl restart systemd-journald

┌──(kali㉿kali)-[~]
└─$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/journal
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/journal'.

┌──(kali㉿kali)-[~]
└─$ sudo /opt/splunkforwarder/bin/splunk restart
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Stopping splunkd...
Shutting down.  Please wait, as this may take a few minutes.

Stopping splunk helpers...

Done.
splunkd.pid doesn't exist...

Splunk> Now with more code!

Checking prerequisites...
        Checking mgmt port [8089]: open
        Checking conf files for problems...
        Done
        Checking default conf files for edits...
        Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-9.3.2-d8bb32809498-linu
x-2.6-x86_64-manifest'
        All installed files intact.
        Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done


┌──(kali㉿kali)-[~]
└─$

┌──(kali㉿kali)-[~]
└─$
```

Right Ctrl
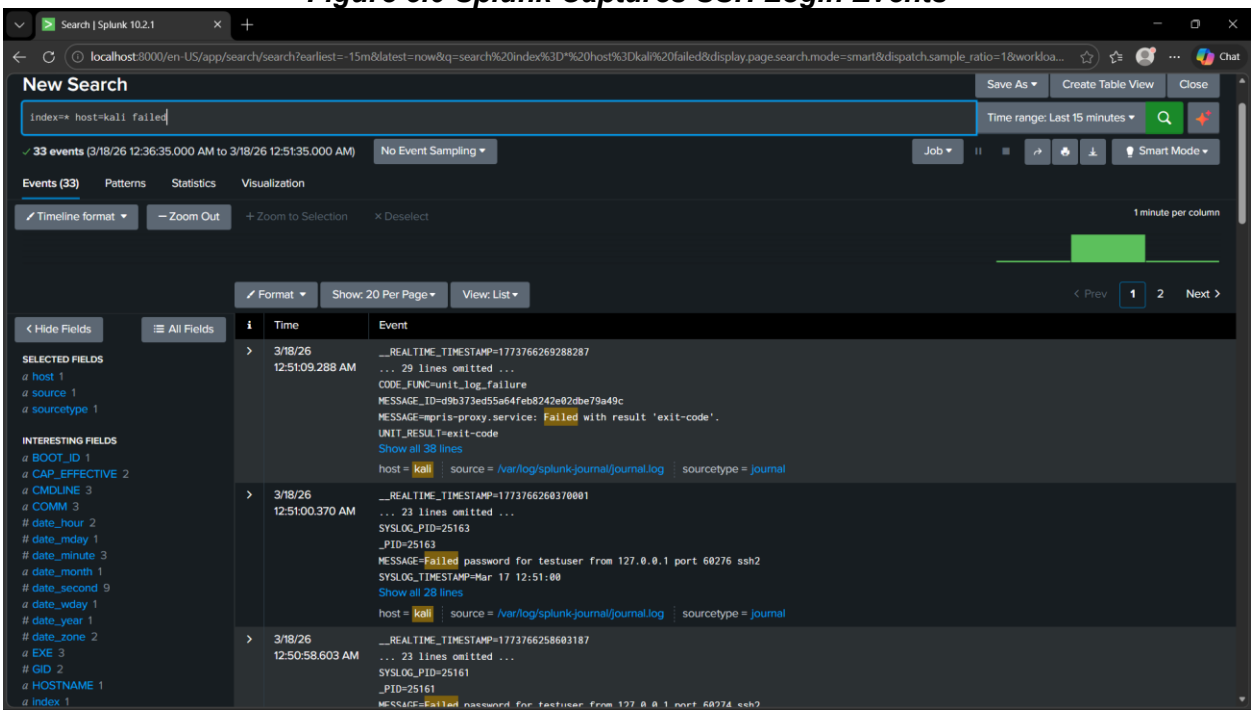
Figure 5.0 Splunk Captures SSH Login Events